

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)		$\frac{\text{[Total] displacement}}{\text{time}}$ Accept rate of change of displacement	1			1		
	(b)	(i)	Speed = $\frac{3.14 \times 1.25}{4}$ [= 0.98 m s ⁻¹]	1			1	1	
		(ii)	Velocity = $\frac{2.5}{4}$ [= 0.63 m s ⁻¹] (1) South (1) accept downwards arrow	1	1		2	1	
		(iii)	Acceleration = rate of change of velocity (or implied) (1) Velocity [constantly] changing (1) since direction is [constantly] changing so Seren is correct (1) Accept leuan is incorrect Award 2 marks max for circular motion stated (1) Centripetal acceleration or acceleration towards centre (1)			3	3		
	(c)		Calculation of time for decelerating e.g. by $x = \frac{(u+v)t}{2}$ so $t = 2.4$ [s] (1) N.B. can be awarded from the graph Appropriate v - t graph vertical axis (0 to 1 or 2) (1) Straight horizontal line on v - t graph from (0,1) to (2,1) and decreasing steadily (4.4,0) (1) ecf on 4.4 if calculated Appropriate displacement-time vertical axis (0 to about 4 m) (ecf) (1) or part of the scale, be tolerant Straight line from (0,0) to (2,2) (1) allow if exceeds 2.2 Smooth curve from (2,2) to (4.4, 3.2) and horizontal at (4.4, 3.2) (1) accept horizontal line continuing beyond 4.4	1	1 1 1 1 1		6	6	
			Question 5 total	4	6	3	13	8	0